



# MULTIMETERS FOR HIGH-END TESTING

**Multimeters—the simplest electrical testing tools—have evolved as high-end electrical and electronic testing equipment to serve complex testing needs. Today, these incorporate a higher level of integration, connectivity and advanced features**

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**M**ultimeters are the most essential and simplest test and measurement (T&M) equipment. These can aptly be called the screwdrivers of the T&M world. Originally designed for technicians to measure mere voltage, current and resistance, they saw a revolution of sorts in the last decade.

“Rapid technological developments have resulted in the need for accurate data and quick results, and instrumentation plays a very decisive role in everyday science,” shares Milind Thakkar, marketing manager, Industrial Group India, Fluke South East Asia. So, he adds, “From industrial electronic installation, maintenance and service to precision measurement and quality control, multimeters have a huge opportunity and market for growth.”

Multimeters, or multiple testers, are the basic testing tools that find diverse use across industries. To meet the needs of every segment, the design of multimeters has evolved from bench-top to portable handheld and lately modular form-factor. Also, their sampling rate, resolution and precision have improved to provide better

measurements and greater usability at an affordable price.

From semiconductor testing and chip designing to aerospace and automotive testing, or precision measurement and data logging in industrial testing, multimeters today fit all the requirements. T&M manufacturers are offering innovatively designed, reliable and high-precision equipment to cater to these segments.

## Integration of new capabilities

To serve the complex testing requirements of today, integrating more parts, devices or features that will complement the use of particular equipment has become a unique selling point for many T&M manufacturers.

Anant K. Waghchoure, market development executive, Rishabh Instruments, says, “Adding new features to the instrument in order to give more value to the user is essential. This has been very useful as the user seeks to extract more from the given product.”

“Major features and functions of a digital multimeter include diode test, low-battery indication, continuity buzzer, data hold, auto power-off, bar-graph display, RS-232 interface

jack, USB interface, function protection, shockproof protection, frequency/duty cycle (0.1-99.9 per cent) and AC frequency response (40-400Hz),” shares Rajesh Joshi, manager, education solutions group, Dynalog India.

Many more measurement capabilities could be added by using accessories and integrating more equipment. Manjunath C.S., business unit head, TTL Technologies, says, “Your digital multimeter can become a thermometer, ampere-meter or even pressure gauge. For example, current clamps can extend the current ranges of digital multimeters and are available for both AC and AC-DC environments.”

Companies have also started selling hybrid or two-in-one kind of equipment. Sadaf Siddique, senior technical marketing engineer from Agilent Technologies, says, “There are equipment that combine multimeters and power supply to address specific testing needs.”

Thakkar adds, “You also have equipment that integrate an insulation tester with a digital multimeter. High-end engineering and safety features make these lightweight tools ideal for field use. They meet the basic multimeter needs of electricians and technicians while giving them an insulation